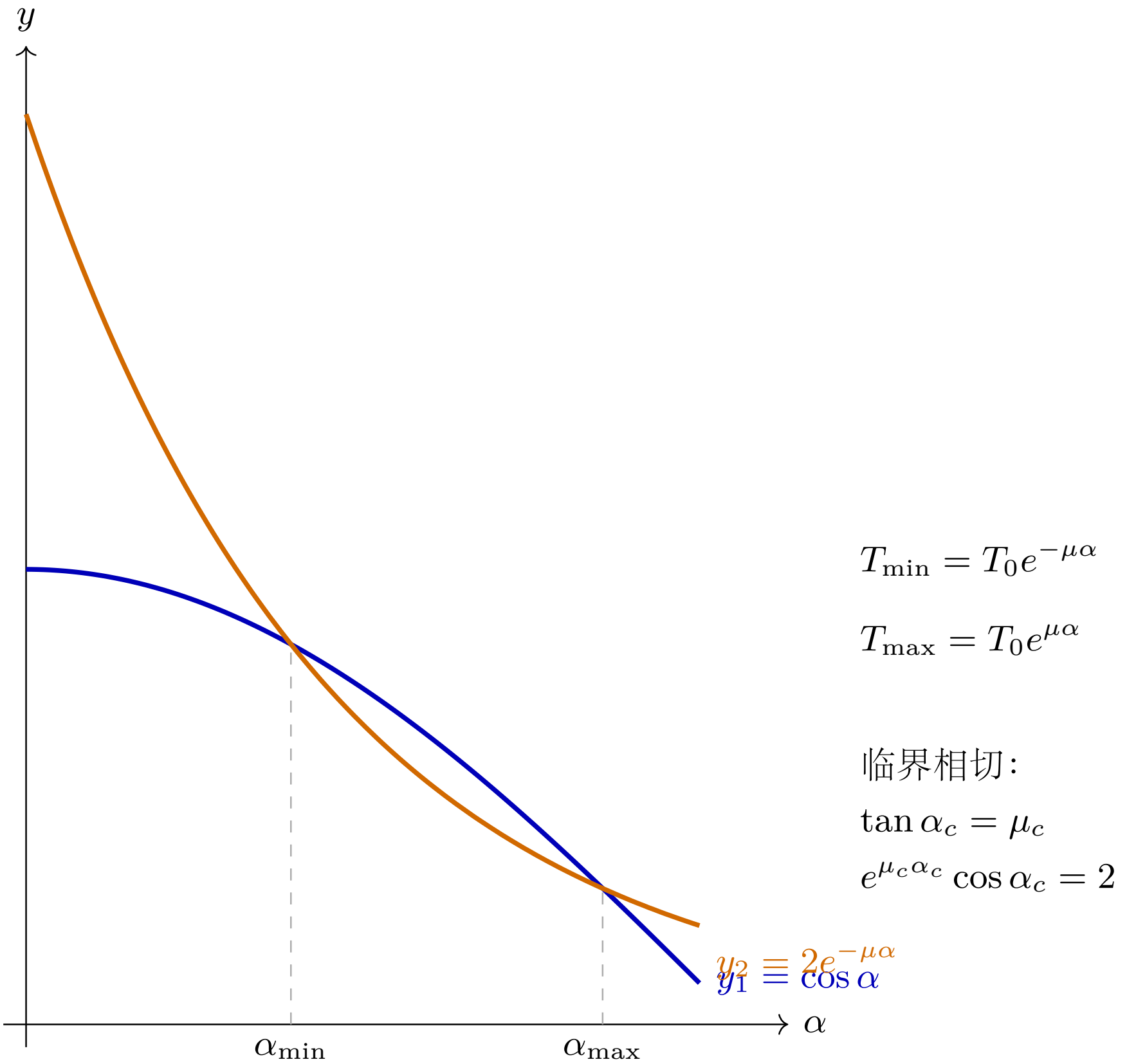


$$T_{\min} = T_0 e^{-\mu\alpha}$$

$$T_{\max} = T_0 e^{\mu\alpha}$$

临界相切：

$$\tan\alpha_c = \mu_c$$

$$e^{\mu_c\alpha_c}\cos\alpha_c = 2$$

上环平衡：$2T\cos\alpha = mg + 2T_0 = 2mg$